



UUM

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STINK3024 - MACHINE LEARNING

GROUP ASSIGNMENT-1

A Descriptive Text Analytics Study “Mapping Technical Skills in AI/ML Job Markets”

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1. Introduction

1.1 Problem Statement

The rapid expansion of Artificial Intelligence (AI) and Machine Learning (ML) across industries has intensified employer demand for workers with specialized technical competencies. However, these competencies are often embedded in unstructured job advertisements, where the same skill may be described in different forms across postings. This makes it difficult to identify which technical skills are genuinely central to the current AI/ML labor market. Verma et al. (2021) show that AI and ML job advertisements contain identifiable skill patterns and that these patterns can be extracted through content analysis to better understand employer demand.

This challenge becomes more significant because AI-related work is evolving faster than traditional job classifications and academic programs can easily adapt. As AI adoption expands across business and industry, the labor market is increasingly shaped by changing task requirements, emerging occupations, and growing expectations for specialized technical ability (Lane & Saint-Martin, 2021; Rawboon et al., 2021). For this reason, there is a clear need for a descriptive analytics approach that can systematically examine current job advertisements and map the skills most strongly associated with AI/ML roles.

1.2 Purpose of Analysis

The purpose of this study is to apply descriptive analytics to AI/ML job advertisements in order to identify the most frequently demanded technical skills, uncover recurring textual patterns, and examine how selected features of job descriptions relate to salary behavior. This is important because online job postings have become a rich source of real-time labor-market information and can be transformed into meaningful labor-market intelligence through data cleansing, normalization, and information extraction techniques (Tzimas et al., 2024).

In addition, natural language processing methods make it possible to analyze large volumes of unstructured job text more systematically than manual review alone. Lukauskas et al. (2023) show that NLP-based analysis of job advertisements can improve understanding of skill demand and support the creation of more structured job profiles. In this project, descriptive analytics is therefore used not only to summarize the dataset, but also to convert

raw job advertisement text into interpretable insights about current AI/ML labor-market expectations.

1.3 Context: The AI/ML Industry (2024–2026)

Between 2024 and 2026, AI and ML technologies have become central to digital transformation across industries, influencing how firms operate, recruit, and define technical work. Rather than affecting only a narrow segment of the technology sector, AI now shapes job roles across finance, healthcare, logistics, services, and other knowledge-intensive industries. Lane and Saint-Martin (2021) argue that the labor-market effects of AI are better understood as a reorganization of tasks and skills within occupations rather than as a simple story of job replacement.

At the same time, the broader Industry 4.0 environment has increased the demand for more advanced and multidisciplinary competencies. Rawboon et al. (2021) emphasize that future employers increasingly seek workers who combine technical knowledge with broader methodological and personal capabilities. In the context of AI/ML hiring, this means employers are often looking for candidates who can work not only with programming and analytics, but also with evolving AI tools, models, and application areas.

1.4 Goals

This project has four main goals. First, it aims to identify the dominant technical themes that appear in AI/ML job descriptions. Second, it seeks to detect recurring word and phrase patterns that characterize current AI-related hiring language. Third, it aims to examine salary patterns through summary statistics and text-based exploration in order to understand whether certain technical signals tend to appear more often in higher-paying roles. Fourth, it aims to generate initial insights that can support later predictive modeling and experimentation.

Overall, the study is descriptive rather than predictive at this stage. Its main objective is to understand the structure of the dataset, reveal meaningful patterns in the job descriptions, and establish an evidence-based foundation for future machine learning work.

1.5 Project Scope

This study focuses on the Global AI/ML Job Market 2026 dataset used in the project. The analysis is limited to descriptive analytics and centers on cleaned job description text together with salary-related variables. The scope of the assignment does not include building a final predictive model; instead, it focuses on text preprocessing, summary statistics, visualization, and exploratory interpretation of labor-market patterns.

This scope is appropriate because descriptive analytics is a necessary first stage before moving into predictive or experimental machine learning tasks. By first identifying recurring skill patterns, salary tendencies, and textual anomalies, the project creates a clearer basis for more advanced modeling in the next stage of the coursework.

2. Related Works

2.1 Previous Studies

The intersection of artificial intelligence, labor market dynamics, and educational alignment has generated a substantial body of research. The 20 reviewed papers can be categorized into three primary themes: AI Market Trends and Employment Patterns, NLP in Recruitment and Job Analysis, and Skill Gap Analysis between Education and Industry.

Theme 1:

AI Market Trends and Employment Patterns The macroeconomic integration of AI has fundamentally disrupted traditional labor markets. The OECD (2021) reports that AI technologies simultaneously automate routine tasks and complement highly skilled labor, leading to both job displacement and the creation of entirely new task categories. Expanding on this, Ros and Loeung (2025) conducted a quantitative study in Cambodia, finding that AI-based task characteristics and information processing capabilities directly influence workforce dynamics and job roles. As AI technologies advance, managing human-AI synergy has become critical; a 2025 review of automation bias emphasizes that mitigating over-reliance on algorithms is essential for effective human-AI decision-making, particularly in high-stakes environments. Furthermore, the rapid adoption of large language models (LLMs) has catalyzed the emergence of entirely novel professions. Vu Quoc and Oppenlaender (2026) highlight the rise of the "Prompt Engineer," revealing that while these

positions remain relatively rare (under 0.5% of sampled postings), they demand a unique, hybrid skill profile blending machine learning expertise with creative prompt design and advanced communication skills.

Theme 2:

Skill Gap Analysis and Educational Alignment A significant portion of the literature focuses on the widening gap between the skills supplied by higher education institutions (HEIs) and the competencies demanded by modern industry. Parvatha (2024) stresses the urgency of optimizing educational curricula using AI-driven analytical frameworks to predict and align with evolving industry needs. In the UK, Jaiswal et al. (2025) quantified this mismatch, demonstrating that while universities successfully supply foundational programming and machine learning skills, they fall significantly short in teaching applied data science, mathematics, and business management skills required by the AI sector. Similarly, Saleema et al. (2025) proposed a machine learning framework to mathematically measure the disparity between HEI student exit profiles and industry job descriptions. Investigating Industry 4.0, Rawboon et al. (2021) identified core future competencies for demanding careers—such as data scientists and robotics engineers—highlighting a critical need for adaptability, active learning, and complex problem-solving.

Theme 3:

NLP and Data Mining in Recruitment The methodological shift toward data-driven recruitment has redefined how organizations source talent. Rouaine et al. (2025) demonstrated that applying data mining in IT recruitment within Morocco significantly optimizes the hiring process, enhances transparency, and enables precise performance prediction. To operationalize these insights, researchers have increasingly relied on Natural Language Processing (NLP) to parse unstructured job advertisements. Tzimas et al. (2024) introduced a pipeline for transforming online job postings into actionable labor-market intelligence, tracking shifting employment types in Greece's tourism industry. Lukauskas et al. (2023) segmented Lithuanian job ads to understand the evolving demand for programming and software competencies. Taking a specialized lens, Grljević and Kecojević (2026) utilized NLP to analyze ICT job advertisements in Serbia from a gender perspective, discovering that male-majority job postings heavily emphasize "hard" engineering terminology, whereas female-majority postings prioritize collaborative and product-oriented "soft" skills. Fleming et al. (2024) applied string detection to over 26,000 engineering job ads to quantify the "skill premium" (salary increases) associated with specific technical tools and professional skills.

Additionally, Verma et al. (2021) utilized content analysis and n-grams to compare the differing requirements of AI versus Machine Learning roles, noting that ML positions are heavily technical while AI roles require more generic, communicative abilities.

2.2 Methods Used

To analyze the unstructured text of job descriptions, the reviewed literature employs a diverse array of advanced text analytics and machine learning methodologies:

- **Vectorization and Word Embeddings (TF-IDF & Scalable Methods):** To process text computationally, researchers must transform unstructured language into numerical vectors through a process known as vectorization. Cahyani and Patasik (2021) compared Term Frequency-Inverse Document Frequency (TF-IDF) with Word2Vec, noting that while Word2Vec captures dense contextual semantic relationships, TF-IDF remains significantly more efficient for high-dimensional text data by prioritizing term importance across large corpora. Although historical approaches often paired Word2Vec with traditional classifiers like **Support Vector Machines (SVM)** to achieve high accuracy in smaller datasets (Rifaldy et al., 2025), modern descriptive analytics on large-scale job portals has shifted toward more scalable architectures. For instance, in real-time recruitment contexts involving thousands of entries, the combination of TF-IDF weighting and cosine similarity is preferred due to its lower computational overhead and its effectiveness in mathematically measuring the alignment between candidate competencies and evolving job requirements (Saleema et al., 2025). This study adopts the TF-IDF approach to ensure robust and efficient feature extraction across the 2026 AI/ML job market dataset.
- **Topic Modeling (LDA & BERTopic):** To discover hidden thematic structures within massive datasets without predefined lists, probabilistic and neural models are used. Labussière and Bol (2026) utilized Latent Dirichlet Allocation (LDA) on millions of UK job postings to identify latent skill profiles, successfully proving that occupations are actually heterogeneous "bundles of skills" rather than monolithic titles. Gurcan et al. (2025) advanced this by applying BERTopic—a Neural Topic Modeling framework leveraging transformer embeddings and UMAP dimensionality reduction—to cluster big data analytics job postings into 26 highly coherent, data-driven skill themes.

- **Deep Learning and Transformers:** State-of-the-art studies treat skill extraction as a complex sequence labeling task. Akkasi (2024) developed an explainable ensemble model combining Convolutional Neural Networks (CNNs), Bidirectional LSTMs (BiLSTM), Conditional Random Fields (CRF), and BERT contextualized embeddings to extract both hard and soft skills from unstructured text with extreme precision.
- **Named Entity Recognition (NER) and Clustering:** Fareri et al. (2021) developed "SkillNER," leveraging rule-based matching alongside Support Vector Machines to automatically map soft skills from scientific texts to the ESCO occupational framework. For segmentation, Lukauskas et al. (2023) evaluated multiple clustering algorithms on vectorized job ads, finding that HDBSCAN (Hierarchical Density-Based Spatial Clustering) and RCBMIDE effectively grouped jobs based on latent technical requirements.

2.3 Gaps Identified

Despite the breadth of the current literature, several critical research gaps and methodological limitations persist:

- **Outdated Data and Lack of Real-Time Analysis:** A vast majority of the studies rely on historical data snapshots (e.g., data from 2019-2021 or the immediate post-COVID era). Because the AI and ML sectors evolve at a blistering pace, these historical analyses fail to capture the real-time demands of the 2026 labor market.
- **Geographic and Sectoral Silos:** Existing studies are highly localized, often restricting their scope to specific countries such as the UK, Morocco, Cambodia, or Serbia (Jaiswal et al., 2025; Rouaine et al., 2025; Ros & Loeung, 2025; Grljević & Kecojević, 2026). This limits the global generalizability of their findings regarding universal AI/ML skill demands.
- **Reliance on Static Taxonomies:** Many methodologies depend on mapping extracted skills to predefined, rigid taxonomies like O*NET or ESCO (Fareri et al., 2021). As noted by Labussière & Bol (2026) and Vu Quoc & Oppenlaender (2026), these top-down frameworks are fundamentally flawed because they cannot rapidly adapt to entirely novel technologies or emergent roles like Prompt Engineering.
- **Imbalanced Treatment of Hard and Soft Skills:** Current NLP models often struggle to balance the extraction of deeply technical hard skills with nuanced interpersonal soft skills. Consequently, some studies explicitly omit soft skills, while others fail to capture the hybrid nature of modern roles.

2.4 Project Alignment

This project, *Mapping Technical Skills in AI/ML Job Markets: A Descriptive Text Analytics Study*, is strategically designed to overcome these identified limitations. Rather than relying on outdated historical datasets, this project utilizes a descriptive text analytics approach to analyze current, real-time data from the 2026 AI/ML job market. By employing advanced NLP techniques to conduct a bottom-up, inductive discovery of skills, the project completely bypasses the constraints of rigid, predefined taxonomies like O*NET or ESCO. This ensures that emerging niche tools and novel competencies are captured exactly as employers articulate them today. Furthermore, aligning with the theoretical framework proposed by Labussi re and Bol (2026), this project treats occupations as fluid "bundles of skills," successfully deconstructing the heterogeneity of AI job titles. By accurately segmenting and quantifying both hard and soft skills without bias, this descriptive analytics study provides the immediate, data-driven intelligence necessary to bridge the gap between academic curricula and the true demands of the contemporary AI labor market.

3. Methodology

3.1 Data Collection

This study utilizes a secondary dataset obtained from an open-source Artificial Intelligence job postings repository available on **Kaggle** (AI Jobs Dataset – 2026 version). The dataset consists of **5,773 job records and 13 initial attributes**. The primary Data Dictionary for this analysis identifies **job_description** as the core textual source for skill extraction and **salary_min** as the key numerical variable for correlation study.

Before preprocessing, the dataset originally contained the following key attributes:

- job_title
- company_name
- location
- employment_type
- experience_level
- company_size

- industry
- job_description
- required_skills
- salary_min
- salary_max
- posting_date
- job_url

These attributes provide structured and unstructured information related to job roles, organizational characteristics, required competencies, and salary ranges.

This dataset is particularly relevant for this study because it contains detailed textual descriptions of technical requirements for AI-related job roles. The **job_description** column includes essential information such as programming languages, machine learning frameworks, analytical tools, and expected responsibilities. These elements allow the dataset to be analyzed using Natural Language Processing (NLP) techniques to explore the relationship between AI skill demand and salary levels.

Therefore, the dataset is suitable for investigating how technical competencies influence compensation patterns in the AI job market.

3.2 Data Preprocessing

Data preprocessing was performed using the **NLTK library** to improve dataset quality and prepare textual information for Natural Language Processing (NLP) analysis. This process included handling missing values and applying a structured text-cleaning pipeline.

First, missing values in the **salary_min** column were handled by removing incomplete records to ensure that salary-related analysis remained accurate and reliable. This step ensures consistency between textual skill features and numerical salary variables.

Next, the **job_description** column underwent several NLP preprocessing steps:

Lowercasing

All text characters were converted into lowercase format to standardize the dataset and avoid duplication of identical words caused by capitalization differences.

Example:

Python Developer → **python developer**

Punctuation Removal

Special characters and punctuation symbols were removed because they do not contribute meaningful semantic information in skill extraction tasks.

Stopword Removal

Common English stopwords such as *the*, *and*, *is*, and *of* were removed using the **standard English stopwords list** from the NLTK library to reduce noise and improve the quality of extracted technical keywords.

Lemmatization

Lemmatization was applied to convert words into their base or dictionary forms. This process was specifically chosen over stemming to maintain terminology consistency, ensuring that technical terms retain their linguistic integrity and accurate meaning within the AI/ML context.

Example:

analyzing → **analyze**

models → **model**

Lemmatization was chosen instead of stemming because it **preserves the semantic meaning of technical terms**, which is essential when analyzing professional job descriptions. Unlike stemming, which may produce incomplete or unnatural root forms, lemmatization generates linguistically correct base words that improve the accuracy of feature extraction in AI skill identification.

After the text was thoroughly cleaned, we proceeded to select the most relevant features for analysis in the following section.

3.3 Feature Engineering

Feature engineering was conducted to select relevant attributes that contribute significantly to identifying relationships between AI skills and salary levels. This phase involves **Feature Extraction** through the creation of the *clean_description* feature, designed to isolate the "core skills" most strongly correlated with compensation.

Two primary features were selected:

Job Description

Feature engineering was conducted to identify the most relevant attributes for analyzing the relationship between AI skill requirements and salary levels.

The primary feature selected in this study is:

job_description (Textual Feature)

The **job_description** column was selected as the main textual feature because it contains comprehensive information about required technical competencies, tools, frameworks, and responsibilities expected from candidates.

Although the dataset includes a **required_skills** column, a large portion of its values contained missing or incomplete information. Therefore, relying solely on this column would reduce analysis reliability.

To address this limitation, the study engineered a new structured textual feature derived from the cleaned version of the job description column (**clean_description**). This engineered feature provides richer and more consistent information for identifying AI-related skill patterns.

Examples of extracted skills include:

- Python
- Machine Learning
- Deep Learning
- SQL
- TensorFlow
- Data Analysis

salary_min (Numerical Feature)

The **salary_min** column was selected as the numerical feature representing the lower bound of the salary range offered. This specific variable was chosen to analyze the **guaranteed minimum salary standard** associated with specific technical competencies.

Combining **clean_description** with **salary_min** enables the study to investigate how specific technical competencies influence salary levels across AI-related occupations.

3.4 Vector Representation

After preprocessing and feature selection, the cleaned textual feature (**clean_description**) was transformed into numerical format using the **Term Frequency–Inverse Document Frequency (TF-IDF)** vectorization technique. **TF-IDF** was preferred over **Bag of Words** because it assigns higher weights to unique and informative terms while penalizing frequent, less descriptive words across the corpus.

- **Term Frequency (TF)**

Measures how frequently a word appears in a document.

- **Inverse Document Frequency (IDF)**

Reduces the weight of commonly occurring words across all documents while increasing the importance of rare but informative words.

By combining these two measurements, TF-IDF assigns higher weights to meaningful technical keywords such as programming languages, frameworks, and analytical tools that frequently appear in AI job descriptions but are not overly common across all postings.

As a result, the cleaned textual data was converted into a **high-dimensional numerical matrix representation**, where each column represents a keyword feature and each row represents a job posting.

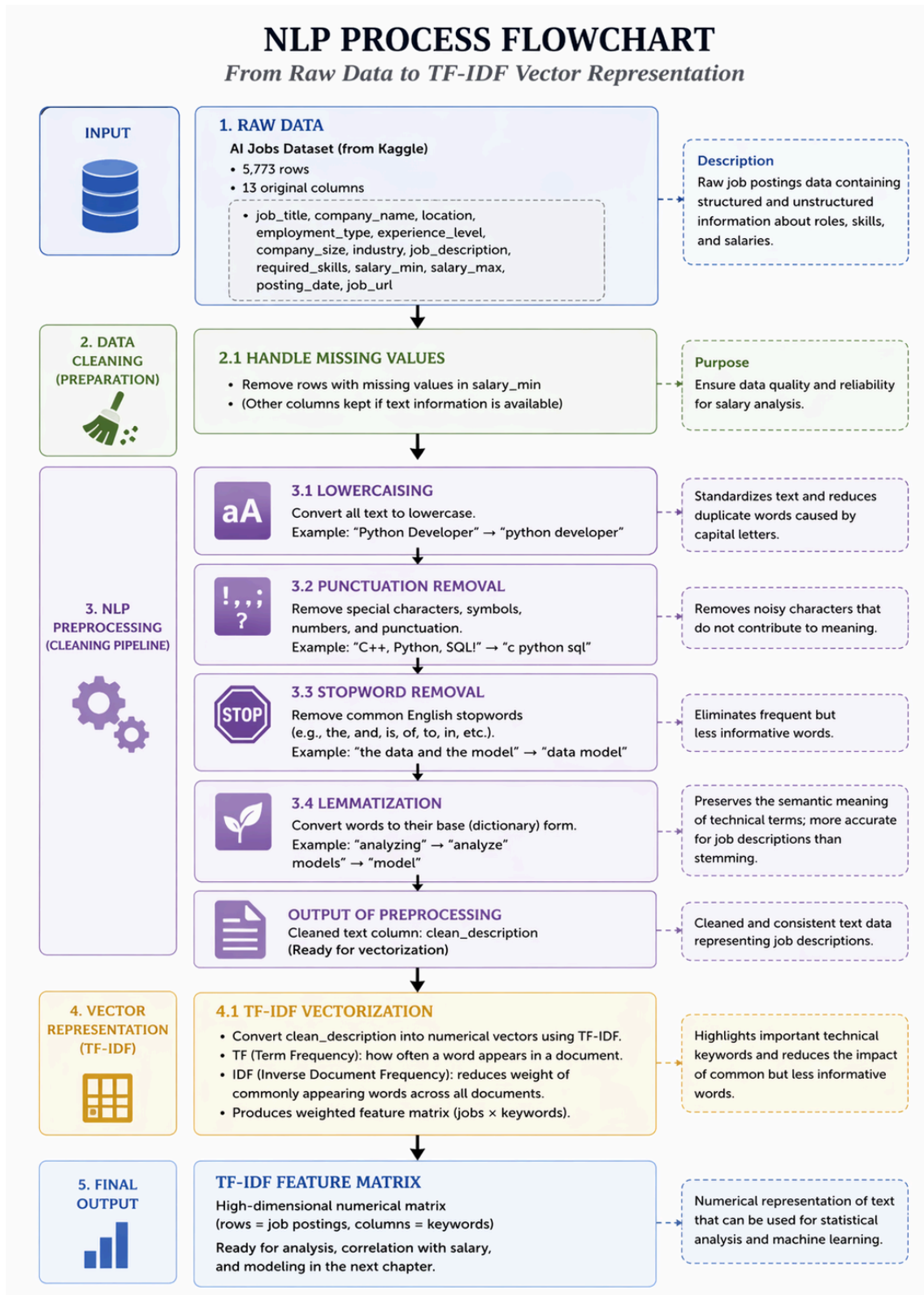


Figure 3.1: Preprocessing Pipeline

4. Data Descriptive Analysis

Conduct exploratory data analysis to understand dataset characteristics:

4.1 Summary Statistics

This section presents the main descriptive statistics for the salary variables used in the analysis, namely `salary_min` and `salary_max`. These variables are central to the study because they provide a direct numerical view of compensation patterns across AI/ML job postings. In the cleaned analytical file, the final dataset contains 5,773 job records, and the salary fields were standardized so that no missing values remained in `salary_min` or `salary_max`.

Table 4.1. Summary Statistics for Salary Variables

Variable	Mean	Median	Standard Deviation	Skewness	Minimum	Maximum
salary_min	106,947.34	103,000.00	36,770.92	1.854	0	431,000
salary_max	113,461.63	103,000.00	45,833.08	2.916	0	800,000

The statistics show that both salary variables are positively skewed, with `salary_max` displaying a stronger right skew than `salary_min`. In practical terms, this means that most job postings are concentrated around a more typical salary range, while a smaller number of postings offer much higher compensation and stretch the distribution upward. The notebook output explicitly reports that `salary_min` has a mean of 106,947.34, a median of 103,000, a standard deviation of 36,770.92, and a skewness of 1.853666, which confirms this right-skewed pattern.

The gap between the mean and the median is also meaningful. Since the mean is higher than the median, the distribution is being pulled upward by relatively high-paying outliers. This indicates that compensation in the AI/ML labor market is not evenly distributed. Instead, there appears to be a concentration of higher-value roles that command a premium above the market center. This interpretation is consistent with OECD discussion that AI's labor-market effects are often tied to task reorganization and differentiated skill demand, rather than uniform outcomes across occupations.

connected to broader organizational work such as problem-solving, product development, and collaboration.

At the same time, the word cloud should be interpreted carefully. A word cloud highlights frequency, but it does not distinguish between general workplace vocabulary and specialized technical requirements. This means that while the figure is useful for showing the dominant language environment of the dataset, it is less precise for identifying actual skills. This limitation is also recognized in labor-market intelligence research, which stresses the need for cleansing, normalization, and more targeted extraction methods when working with free-text job postings.

N-Gram Interpretation

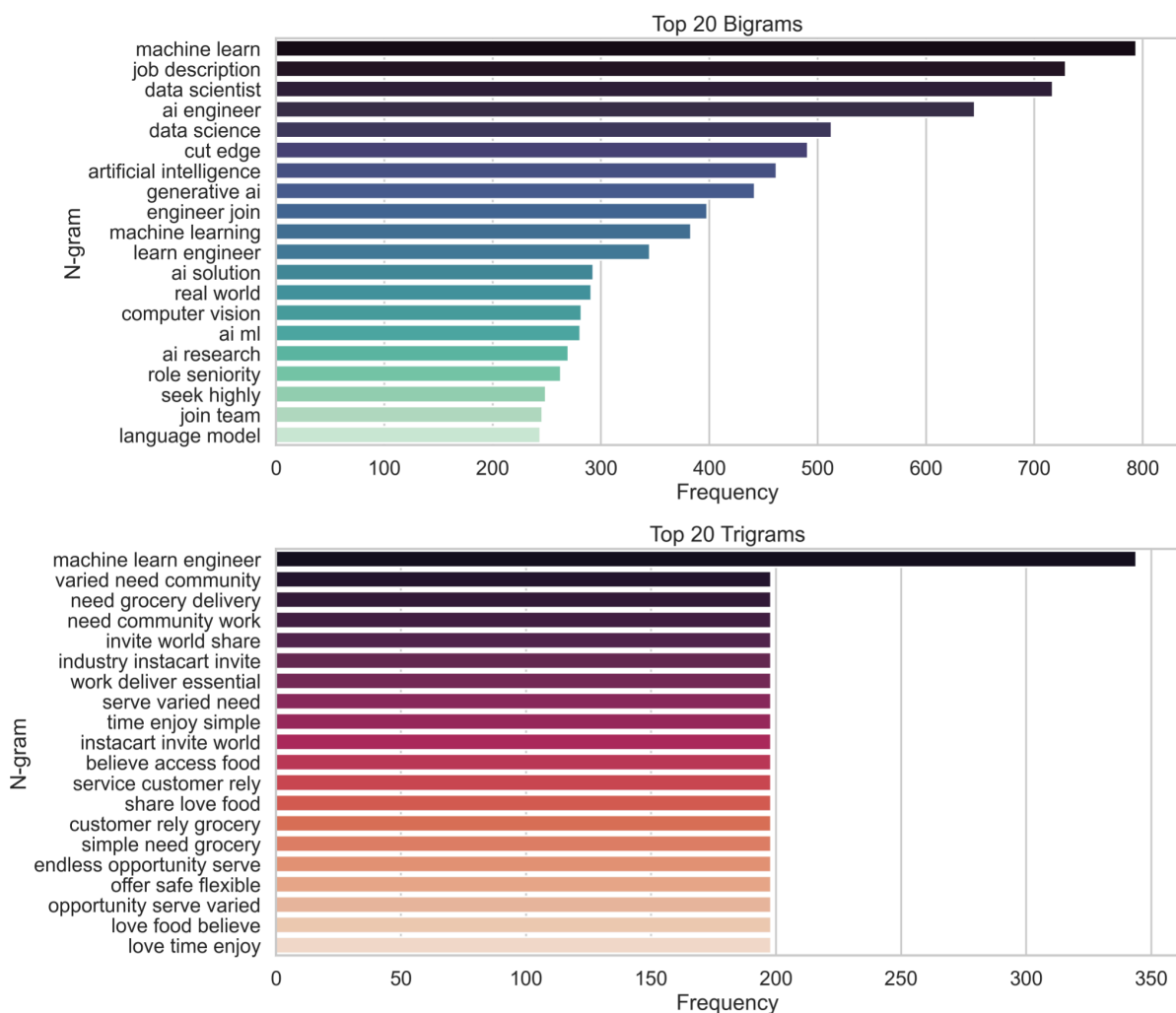


Figure 4.2: Top 20 Bigrams and Trigrams in AI/ML Job Descriptions

The n-gram analysis gives a clearer picture of the technical themes than the word cloud. The bigrams include highly meaningful phrases such as machine learn, data scientist, ai engineer, data science, artificial intelligence, generative ai, computer vision, and language model. These phrases strongly suggest that the dataset reflects current employer demand for both foundational machine learning roles and newer AI specializations.

This pattern aligns well with existing literature showing that job advertisements can be mined to reveal evolving skill requirements and job profiles through NLP-based analysis. Lukauskas et al. (2023) demonstrate that NLP and clustering can help identify emerging skill demand from large-scale job advertisements, while Tzimas et al. (2024) emphasize that online job postings can be transformed into actionable labor-market intelligence when relevant information is extracted from the description text.

However, the trigram results also reveal an important weakness in the corpus. Several trigrams appear to reflect repeated employer boilerplate or company-specific phrasing rather than transferable technical competencies. Phrases such as need grocery delivery, service customer rely, and similar repeated structures suggest that some advertisements contain promotional or template language that was preserved even after cleaning. This is an important analytical finding because it shows that real-world job posting data often mixes genuine skill signals with repetitive noise. As Tzimas et al. (2024) note, missing values, inconsistent language, and noisy free-text content are common issues in job-posting analysis and need to be handled carefully in interpretation.

Salary Boxplot Interpretation

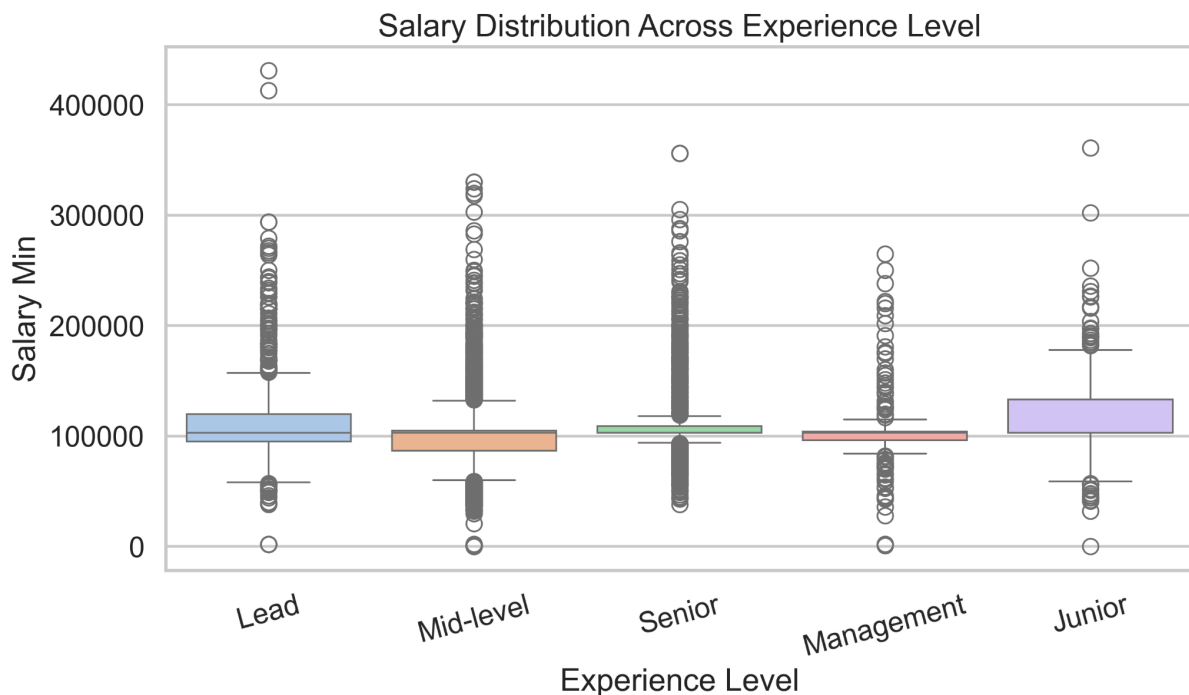


Figure 4.3: Salary Distribution Across Experience Level

The salary boxplot compares `salary_min` across different experience levels, including Junior, Mid-level, Senior, Lead, and Management roles. The figure suggests that salaries are dispersed across all categories, but the variation becomes particularly noticeable in advanced positions. Lead and Senior roles show a broader spread and more extreme upper-end outliers, which implies that higher-level positions are more likely to capture salary premiums.

Even so, the plot does not suggest a simple, perfectly ordered salary ladder. Junior roles also show meaningful variability, and some upper-level categories overlap considerably around the center of the distribution. This indicates that experience level alone does not fully explain compensation. Instead, salary appears to reflect a combination of seniority, specialization, organizational context, and perhaps the presence of niche AI-related requirements. This interpretation is consistent with OECD findings that AI tends to reorganize tasks within occupations and create differentiated value across roles, rather than producing a single uniform pattern across the labor market.

A further observation is that upper-end outliers appear in nearly every group. This suggests that premium salaries are not restricted to only one job category but may emerge wherever

employers require unusually strong technical depth, strategic responsibility, or scarce skill combinations. For the purpose of this assignment, the boxplot therefore supports the conclusion that AI/ML salaries are heterogeneous and should be analyzed with caution rather than summarized by a single average alone.

4.3 Correlation Analysis

Heatmap and TF-IDF Correlation Interpretation

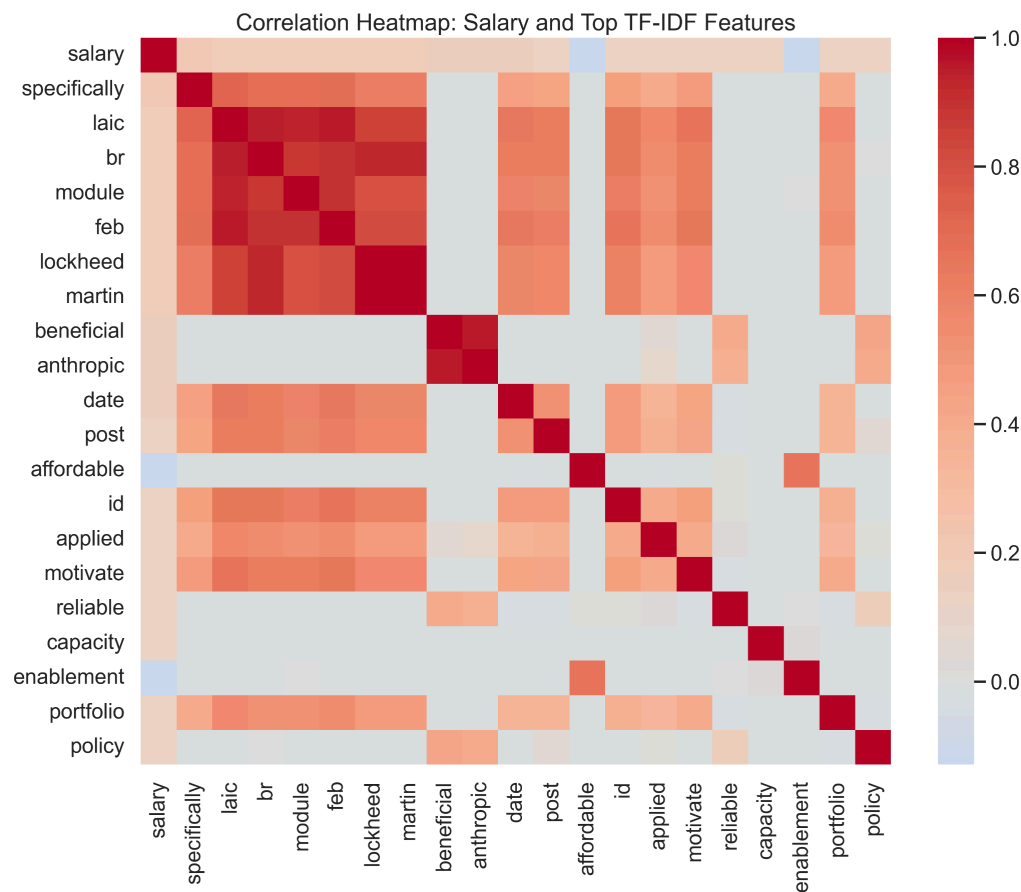


Figure 4.4: Correlation Heatmap of Salary and Top TF-IDF Features

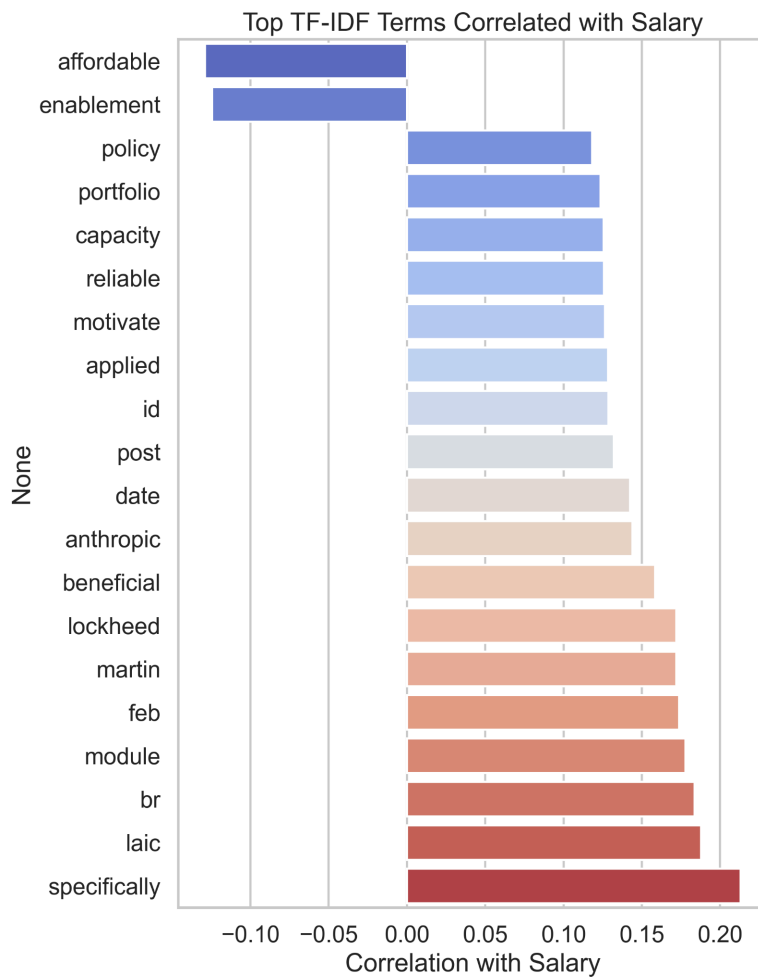


Figure 4.5: Top TF-IDF Terms Correlated with Salary

The correlation analysis was used to examine whether certain TF-IDF terms tend to appear more often in postings associated with higher salary values. The bar chart shows that terms such as specifically, laic, br, module, feb, lockheed, and martin are among the most positively correlated with salary, while affordable and enablement show slight negative associations.

The most important point here is not the size of the correlations, but their nature. Several highly ranked terms do not appear to be generalizable technical skills. Instead, they seem to reflect company names, contextual metadata, posting fragments, or highly specific wording. For example, lockheed and martin likely reflect employer-specific text rather than transferable technical capabilities. This is a strong indication that correlation in job text can capture multiple types of information at once, including skill content, organizational identity, and advertisement structure.

This is exactly why TF-IDF remains useful but must be interpreted carefully. As Cahyani and Patasik (2021) explain, TF-IDF is valuable because it gives greater importance to distinctive terms and reduces the dominance of overly frequent words. At the same time, labor-market studies based on job advertisements repeatedly stress that free-text job data can contain noise, duplicated language, and inconsistent expressions, all of which can affect the quality of extracted features.

Multicollinearity Note

A multicollinearity check was considered in this stage because TF-IDF features often co-occur and may carry overlapping information. In text-based datasets, several terms can repeatedly appear together across similar advertisements, which creates redundancy and makes it difficult to interpret each feature as a separate signal. For this reason, the analysis focused on the **most distinctive and interpretable** terms rather than assuming that every correlated TF-IDF feature represented an independent skill.

This step improves the descriptive quality of the analysis. It reduces the risk of over-interpreting terms that simply travel together because of template wording, company-specific language, or repetitive posting structures. The need for careful preprocessing, dimensionality reduction, and feature simplification is also emphasized in job-advertisement NLP research, which notes that unstructured posting text produces high-dimensional feature spaces and requires disciplined data preparation before reliable interpretation.

4.4 Initial Insights

Several important insights emerged from the descriptive analysis. First, the AI/ML job market represented in this dataset is clearly shaped by both **core technical roles** and **newer AI specializations**. The n-grams show strong emphasis on **machine learning**, **data scientist**, **AI engineer**, **computer vision**, **language model**, and **generative AI**, which indicates that employer demand is moving beyond traditional analytics into more advanced AI application areas.

Second, the salary analysis suggests that compensation is unevenly distributed across the market. The positive skew in both salary variables, together with the wide spread across experience levels, implies that salary premiums are concentrated in a smaller number of postings. This suggests that the labor market rewards particular combinations of specialization, responsibility, and domain relevance rather than simply rewarding years of experience in a linear way.

Third, one of the most useful “aha” moments in the project is that not every frequent or correlated term is actually a skill. The trigram and correlation outputs make this very clear. Repeated company language, promotional wording, and employer-specific references can survive preprocessing and become visible in the descriptive outputs. This is not a failure of the analysis; rather, it is a meaningful result in itself, because it shows where future cleaning and feature engineering must improve.

Finally, these descriptive findings are directly useful for the next machine learning assignment. The dominant themes identified here can guide feature selection for future tasks such as salary prediction, job-role classification, or skill-demand forecasting. In particular, future models should preserve informative AI-related indicators, such as language-model, computer-vision, and generative-AI signals, while down-weighting repetitive boilerplate and employer-specific noise. This is consistent with the broader literature, which treats descriptive analysis of job advertisements as a foundation for more structured labor-market intelligence and later machine-learning applications.

4.5 Conclusion & Recommendations

This study provides a descriptive view of how technical skill demand is currently expressed in the Global AI/ML Job Market 2026 dataset. Across the statistical summaries, text-based visualizations, and correlation analysis, the results indicate that employer demand is centered on a combination of foundational and emerging AI competencies. The most visible patterns in the dataset point to machine learning, data science, AI engineering, computer vision, language-model-related work, and generative AI themes. In practical terms, this suggests that the market is no longer focused only on general analytical ability, but increasingly values

specialized AI capabilities, including modern model development frameworks and LLM-oriented workflows.

The findings also show that salary is not distributed uniformly across the AI/ML labor market. Instead, the salary variables are positively skewed, meaning that while many positions cluster around a typical salary range, a smaller group of roles receives substantially higher compensation. This pattern suggests that premium salaries are more likely attached to specialized technical combinations, advanced responsibility, or niche high-value roles. At the same time, the descriptive analysis revealed that job advertisement text contains both meaningful skill signals and a degree of repetitive or employer-specific wording. This is an important result because it shows that descriptive analytics is useful not only for identifying trends, but also for detecting data quality issues that could affect later modeling stages.

Based on these findings, this study recommends that future machine learning work should build directly on the descriptive insights generated here. The dominant technical themes identified in this chapter can be translated into candidate features for later tasks such as salary prediction, job-role classification, or skill-demand forecasting. In particular, future experimentation should prioritize informative textual features linked to specialized AI skills, reduce the influence of repetitive non-skill phrases, and test whether refined TF-IDF-based representations can improve predictive performance. In this way, the current descriptive analytics stage serves as an evidence-based foundation for more advanced machine learning modeling in the next assignment.

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